Angle chasing and polygons

L2-GE1 Level 2 Intermediate · Geometry

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How to find angles quickly using parallel lines, isosceles triangles, polygon angle sums and the basic circle theorems.
By the end you can chase angles through parallel lines and isosceles triangles, naming one unknown when stuck; use polygon angle sums and the angle at the centre theorem with its consequences; spot hidden circles and use the circumcircle of a regular polygon to read off angles.
Assumes: L1-GE1 (L1-GE1). **Contests:** IMC, Grey Kangaroo, Pink Kangaroo, Cayley, Hamilton, JMO, AMC 10. **Time:** about 150 minutes.

1 Lesson

In triangle ABC the angle at A is 50°, and the bisectors of the angles at B and C meet at I. What is $\angle BIC$?

We are not told $\angle B$ or $\angle C$, and they cannot be found, so call them $2b$ and $2c$. The angle sum of ABC gives $2b + 2c = 130^\circ$, so $b + c = 65^\circ$. In triangle BIC the angles at B and C are b and c, so $\angle BIC = 180^\circ - 65^\circ = 115^\circ$. Note that b and c were never found; only $b + c$ was needed. The same argument shows $\angle BIC = 90^\circ + \frac{1}{2}\angle A$ for every triangle.

This is angle chasing: name the unknown angles, apply the basic facts, and continue until one angle has been written in two ways. This unit collects the facts you need.

1.1 Basic facts

Angles on a straight line add to 180°, angles round a point add to 360°, and vertically opposite angles are equal. When a line crosses two parallel lines, *alternate* angles (the Z shape) are equal, *corresponding* angles (the F shape) are equal, and *co-interior* angles (the C shape) add to 180°. Each of these also works backwards: if alternate angles are equal, the lines are parallel.

Triangle facts.
The angles of a triangle add to 180°. The exterior angle at one vertex equals the sum of the interior angles at the other two vertices.

To see why, draw the line through A parallel to BC. It makes alternate angles equal to $\angle B$ and $\angle C$ with AB and AC, and together with $\angle A$ these fill a straight line. The exterior angle at C is $180^\circ - \angle C$, which is therefore $\angle A + \angle B$. Use the exterior angle fact directly, rather than going through the angle sum.

1.2 Isosceles triangles

If two sides of a triangle are equal, the angles opposite them are equal, and conversely. If the apex angle is θ, each base angle is $90^\circ - \frac{1}{2}\theta$. Most contest angle problems depend on isosceles

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triangles, so the first thing to do with any diagram is to look for equal lengths. They come from three places: lengths you are told are equal, sides of regular polygons (and of the squares and equilateral triangles attached to them), and radii of a circle.

When a problem only says that an isosceles triangle has an angle of 40°, there are two triangles: $40^\circ, 70^\circ, 70^\circ$ and $40^\circ, 40^\circ, 100^\circ$. Check both.

1.3 Polygons

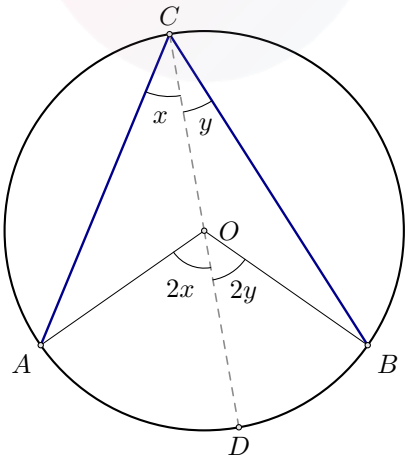
Polygon angles.
The interior angles of a convex n-sided polygon add to $(n-2)\times 180^\circ$, and its exterior angles add to 360°. So each interior angle of a regular n-gon is $180^\circ - \dfrac{360^\circ}{n}$.

For the first fact, join one vertex to every other vertex: this cuts the polygon into $n-2$ triangles, and their angles together make up the interior angles. For the second, walk round the boundary. At each corner you turn through the exterior angle, and after one lap you face the way you started, having made one full turn. Know the regular values: $60^\circ, 90^\circ, 108^\circ, 120^\circ, 135^\circ, 140^\circ, 144^\circ, 150^\circ$ for $n = 3, 4, 5, 6, 8, 9, 10, 12$.

1.4 Circle theorems

Angle at the centre.
If A, B, C lie on a circle with centre O, and C is on the major arc AB, then $\angle AOB = 2\angle ACB$.

Draw the diameter CD. Since $OA = OC$, triangle OAC is isosceles, so if $\angle OCA = x$ then $\angle OAC = x$ as well, and the exterior angle $\angle AOD$ equals $2x$. In the same way $\angle OCB = y$ gives $\angle DOB = 2y$. Adding, $\angle AOB = 2x + 2y = 2\angle ACB$. If O happens to lie outside triangle ABC, subtract instead of adding and the same argument works.



The other standard facts follow from this, with one extra fact needed for tangents.

- **Same segment.** Angles standing on the same chord, on the same side of it, are equal: each is half the same angle at the centre.

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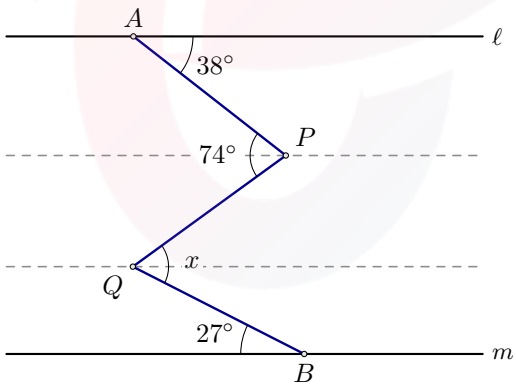
- **Semicircle.** The angle in a semicircle is $90°$, because the angle at the centre is a straight angle.
- **Cyclic quadrilateral.** Opposite angles add to $180°$, because the two angles at the centre make a full turn.
- **Tangents.** A tangent is perpendicular to the radius at its point of contact. The angle between a tangent and a chord equals the angle in the opposite segment (the alternate segment theorem): if AD is the diameter from the point of contact A, both angles equal $90° - \angle BAD$, where B is the other end of the chord.

Regular polygons are cyclic.

The vertices of a regular n-gon lie on a circle, and each side subtends $\dfrac{360°}{n}$ at the centre. So each side subtends $\dfrac{180°}{n}$ at every other vertex, and any angle formed by three vertices is $\dfrac{180°}{n}$ times the number of sides on the arc it stands on.

1.5 The method

To chase angles, we have the following method. Mark every angle you know on the diagram. Look for equal lengths before anything else and mark the isosceles triangles they create. If you get stuck, call one well-chosen angle x (usually the one everything else depends on) and write the other angles in terms of it. You are finished when one angle has been expressed in two different ways. If three equal lengths come out of one point, draw the circle they define.



Example 1.

Lines ℓ and m are parallel. A path goes from A on ℓ to P, then to Q, then to B on m, as shown. The angle between ℓ and AP is $38°$, $\angle APQ = 74°$, and the angle between QB and m is $27°$. Find $x = \angle PQB$.

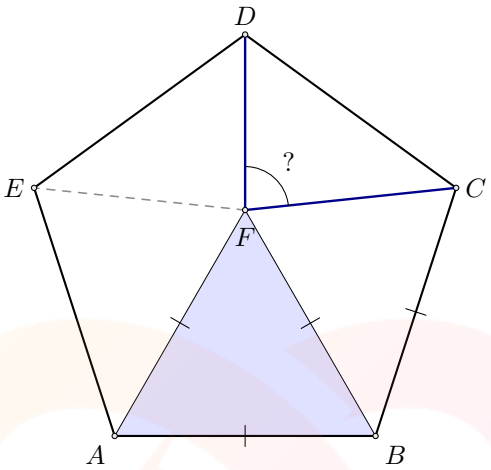
Solution.

We could extend AP to meet m and look for triangles, but the only facts that connect angles at different points here are the parallel line facts. So draw the dashed lines through P and Q parallel to ℓ (one parallel through each bend).

The dashed line through P splits the $74°$ angle into two parts. The upper part and the $38°$ angle at A are alternate angles between parallel lines, so the upper part is $38°$ and the lower part is $74° - 38° = 36°$. At Q the dashed line splits x into two parts. The upper part

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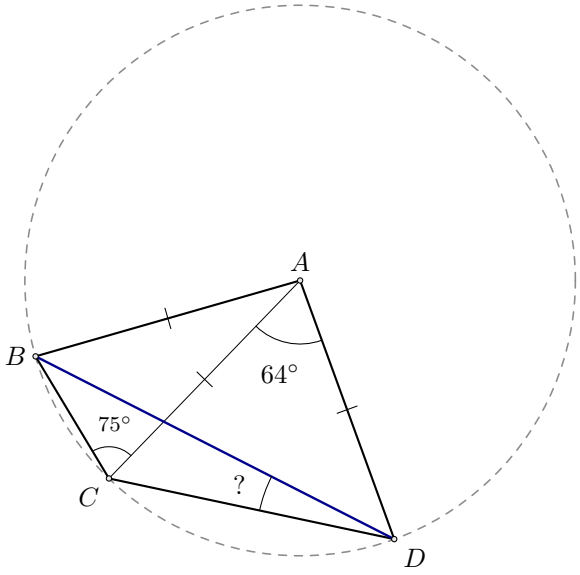
is alternate to that 36° (the segment PQ crosses two parallel lines), and the lower part is alternate to the 27° at B. So $x = 36^\circ + 27^\circ = 63^\circ$.
The same argument proves a rule worth remembering: in a zigzag between two parallel lines, the angles opening to the right add up to the same total as the angles opening to the left. Here $38^\circ + x = 74^\circ + 27^\circ$, which gives $x = 63^\circ$ in one line.



Example 2.
$ABCDE$ is a regular pentagon, and ABF is an equilateral triangle drawn inside it. Find $\angle CFD$.

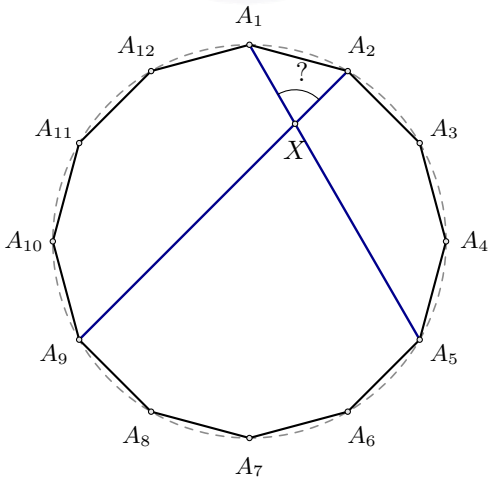
Solution.
The angle we want is in triangle CFD, about which we know nothing directly. But there are many equal lengths: the five sides of the pentagon, and $FA = FB = AB$. In particular $BF = BC$, so triangle BFC is isosceles, and this is the step that starts the chase.
The interior angle of the pentagon is 108°, so $\angle FBC = 108^\circ - 60^\circ = 48^\circ$, and the base angles of triangle BFC are $\angle BFC = \angle BCF = \frac{1}{2}(180^\circ - 48^\circ) = 66^\circ$. Hence $\angle FCD = 108^\circ - 66^\circ = 42^\circ$.
We need one more angle of triangle CFD. The whole figure is symmetric about the line through D and the midpoint of AB, and F lies on this line because $FA = FB$. So FD bisects $\angle CDE$, giving $\angle FDC = 54^\circ$. Therefore $\angle CFD = 180^\circ - 42^\circ - 54^\circ = 84^\circ$.
We check it another way. By the same isosceles argument $\angle AFE = 66^\circ$, and by symmetry $\angle CFD = \angle DFE$. The angles round F give $60^\circ + 66^\circ + 66^\circ + 2\angle CFD = 360^\circ$, so $\angle CFD = 84^\circ$ again.

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Example 3.
$ABCD$ is a convex quadrilateral with $AB = AC = AD$. Given that $\angle ACB = 75°$ and $\angle CAD = 64°$, find $\angle BDC$.

Solution.
Three equal lengths give three isosceles triangles, so a direct chase works. Triangle ABC has $AB = AC$, so $\angle ABC = 75°$ and $\angle BAC = 30°$. Then $\angle BAD = 30° + 64° = 94°$, and the isosceles triangle ABD gives $\angle ADB = \frac{1}{2}(180° - 94°) = 43°$. The isosceles triangle ACD gives $\angle ADC = \frac{1}{2}(180° - 64°) = 58°$. So $\angle BDC = 58° - 43° = 15°$.
Note that the $64°$ cancelled. When a given value does not affect the answer, look for a shorter argument. Since $AB = AC = AD$, the points B, C, D lie on a circle with centre A. The angle $\angle BDC$ stands on the chord BC, which subtends $\angle BAC = 30°$ at the centre, and D is on the major arc. So $\angle BDC = 15°$ immediately, and this explains why D could slide anywhere on that arc without changing the answer.



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Example 4.
$A_1A_2\ldots A_{12}$ is a regular 12-gon. The diagonals A_1A_5 and A_2A_9 meet at X. Find $\angle A_1XA_2$.

Solution.
Interior angles of 150° and isosceles triangles would work, but slowly, especially for $\angle A_1A_2A_9$. Instead, note that all twelve vertices lie on a circle, so every angle between vertices is an angle at the circumference.
Each side subtends $\frac{360^\circ}{12} = 30^\circ$ at the centre, so it subtends 15° at any other vertex. The angle $\angle A_5A_1A_2$ stands on the arc $A_2A_3A_4A_5$, which contains three sides, so it is $3 \times 15^\circ = 45^\circ$. The angle $\angle A_1A_2A_9$ stands on the arc from A_9 through A_{10}, A_{11}, A_{12} to A_1, which contains four sides, so it is 60°. In triangle A_1A_2X we get $\angle A_1XA_2 = 180^\circ - 45^\circ - 60^\circ = 75^\circ$.
The same argument proves a general fact: two chords crossing inside a circle make an angle equal to half the sum of the central angles of the arcs they cut off, here $\frac{1}{2}(30^\circ + 120^\circ) = 75^\circ$.

Common traps.

- Trusting the picture. Contest diagrams are often not to scale. Lines are parallel, or a point is the centre, only if you are told so or can prove it.
- Forgetting the second case. An isosceles triangle with a 40° angle could have it at the apex or at the base.
- Using the centre theorem with the wrong arc. If C is on the minor arc AB, it is the reflex angle AOB that equals $2\angle ACB$.
- Applying "same segment" to points on opposite sides of a chord. Those angles add to 180° (a cyclic quadrilateral); they are not equal.
- Confusing the sum $(n-2) \times 180^\circ$ with a single interior angle, or interior angles with exterior ones.

1.6 How to spot this in a contest

If a question gives angles and asks for an angle, it is a chase. Before writing anything, scan for equal lengths (given ones, sides of regular polygons, radii) and mark the isosceles triangles. If the answer seems to need information you do not have, name the unknowns and expect them to cancel, as in the opening problem. If a given number seems irrelevant, or several equal segments leave one point, look for a circle. For regular polygons with many sides, almost always use the circumcircle and the rule of $\frac{180^\circ}{n}$ per side. A careful sketch can check an answer, but it never replaces the reasoning.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆ Each interior angle of a regular polygon is seven times each of its exterior angles. How many sides does the polygon have?

Problem 2 ⋆⋆ $ABCDEF$ is a regular hexagon and $ABPQR$ is a regular pentagon. They share the side AB and lie on opposite sides of it. Find $\angle AFR$.

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Problem 3 ⋆⋆ Point D lies on side AC of triangle ABC, and $AB = BD = DC$. Given that $\angle ABC = 102°$, find $\angle BAC$.

Problem 4 ⋆⋆⋆ $ABCD$ is a cyclic quadrilateral with $AB = AD$ and $\angle BCD = 110°$. Its diagonals meet at E, and $\angle AEB = 80°$. Find $\angle CAD$.

Problem 5 ⋆⋆⋆ In a regular n-gon $A_1A_2A_3\ldots A_n$, the diagonals A_1A_4 and A_2A_6 meet at X, and $\angle A_1XA_2 = 27°$. Find n.

Problem 6 ⋆⋆⋆⋆ Seven points $A_1, A_2, \ldots, A_7$ lie on a circle in that order, not necessarily equally spaced. Each point is joined to the two points two places away from it round the circle, so A_1 is joined to A_3 and to A_6, and so on. This draws a seven-pointed star. Find the sum of the seven angles at the tips of the star, $\angle A_6A_1A_3 + \angle A_7A_2A_4 + \angle A_1A_3A_5 + \cdots + \angle A_5A_7A_2$.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **AMC 10B 2010 Problem 6** ⋆
   Two radii make an isosceles triangle.
2. **UKMT IMC 2018 Question 4** ⋆
   Write every angle of the kite in terms of x and add.
3. **AMC 10A 2007 Problem 8** ⋆
   Base angles of two isosceles triangles on the same base.
4. **AMC 10B 2007 Problem 7** ⋆
   Split the pentagon into shapes you recognise.
5. **UKMT IMC 2017 Question 17** ⋆⋆
   Regular pentagon angles plus a quadrilateral angle sum.
6. **AMC 10B 2009 Problem 9** ⋆⋆
   A chain of isosceles triangles: name one angle and chase.
7. **UKMT IMC 2018 Question 18** ⋆⋆
   Combine the angles of a square, a regular pentagon and an equilateral triangle.
8. **AMC 10B 2025 Problem 5** ⋆⋆
   The angle at B is obtuse: which arc does it stand on?
9. **AMC 10A 2019 Problem 13** ⋆⋆⋆
   Angles in a semicircle, then an isosceles triangle.
10. **UKMT JMO 2026 Question 5** ⋆⋆⋆
    Compute the angle at F in two different ways.
11. **UKMT Cayley 2026 Question 4** ⋆⋆⋆
    Call one angle theta and chase both base angles of triangle DEF.
12. **AMC 10B 2008 Problem 24** ⋆⋆⋆⋆
    Isosceles triangles first; then an extra point making an equilateral triangle helps.

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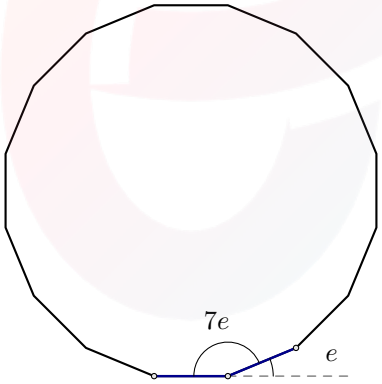
4 Hints

1. At each vertex the interior and exterior angles add to $180°$, and you know what all the exterior angles add to.
2. Work out the angle FAR first, then look at the lengths AF and AR.
3. Call $\angle ACB = x$ and write every other angle in terms of x, using the exterior angle of triangle BDC.
4. Opposite angles of $ABCD$ give $\angle BAD$. Then use the isosceles triangle ABD and angles in the same segment.
5. Put the polygon in its circumcircle. Each side subtends $\frac{180°}{n}$ at any other vertex.
6. Each tip angle is an angle at the circumference. Which arc does it stand on, and how many times is each small arc used?

5 Solutions

Problem 1

Each interior angle of a regular polygon is seven times each of its exterior angles. How many sides does the polygon have?



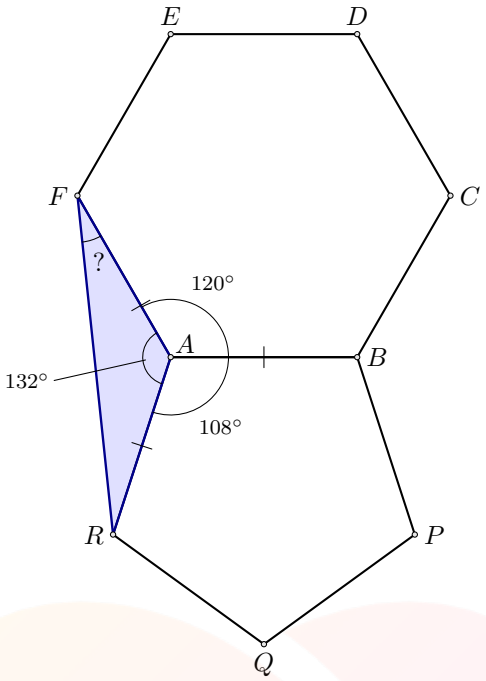
Let each exterior angle be e. Then each interior angle is $7e$, and since they lie on a straight line, $e + 7e = 180°$, so $e = 22.5°$. The exterior angles of any polygon add to $360°$, so the number of sides is $360 \div 22.5 = 16$.

Answer: 16

Problem 2

$ABCDEF$ is a regular hexagon and $ABPQR$ is a regular pentagon. They share the side AB and lie on opposite sides of it. Find $\angle AFR$.

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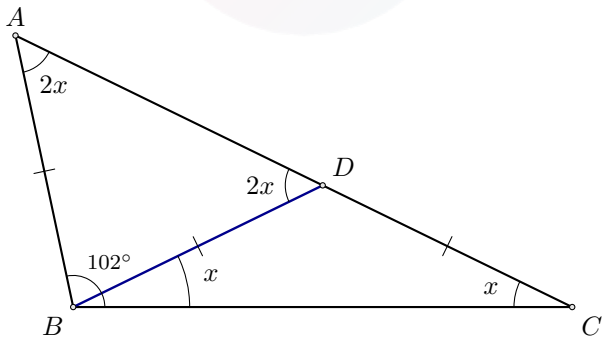
At A the hexagon has interior angle $\angle FAB = 120^\circ$ and the pentagon has interior angle $\angle BAR = 108^\circ$. These two angles lie on opposite sides of AB, so the remaining angle round A is $\angle FAR = 360^\circ - 120^\circ - 108^\circ = 132^\circ$.

Both polygons have all their sides equal to AB, so $AF = AB = AR$ and triangle AFR is isosceles with apex A. Its base angles are $\angle AFR = \angle ARF = \frac{1}{2}(180^\circ - 132^\circ) = 24^\circ$.

Answer: 24°

Problem 3

Point D lies on side AC of triangle ABC, and $AB = BD = DC$. Given that $\angle ABC = 102^\circ$, find $\angle BAC$.



Let $\angle ACB = x$. Since $DB = DC$, triangle BDC is isosceles and $\angle DBC = x$. The angle $\angle BDA$ is an exterior angle of triangle BDC, so $\angle BDA = 2x$.

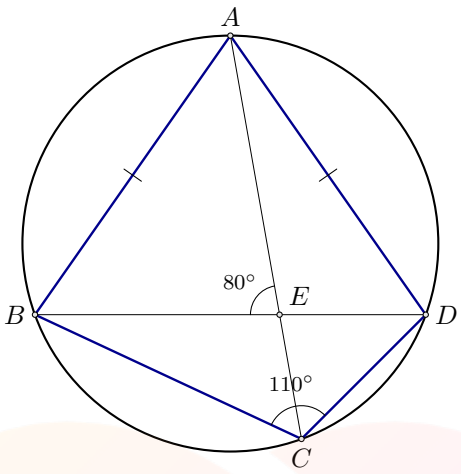
Since $BA = BD$, triangle ABD is isosceles and $\angle BAD = \angle BDA = 2x$, which leaves $\angle ABD = 180^\circ - 4x$. Now $\angle ABC = \angle ABD + \angle DBC = 180^\circ - 3x$. Setting this equal to 102° gives $x = 26^\circ$, so $\angle BAC = 2x = 52^\circ$. As a check, the angles of ABC are 52°, 102° and 26°, which add to 180°.

Answer: 52°

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Problem 4

$ABCD$ is a cyclic quadrilateral with $AB = AD$ and $\angle BCD = 110°$. Its diagonals meet at E, and $\angle AEB = 80°$. Find $\angle CAD$.



Opposite angles of a cyclic quadrilateral add to $180°$, so $\angle BAD = 70°$. Triangle ABD is isosceles with $AB = AD$, so $\angle ABD = \angle ADB = 55°$.

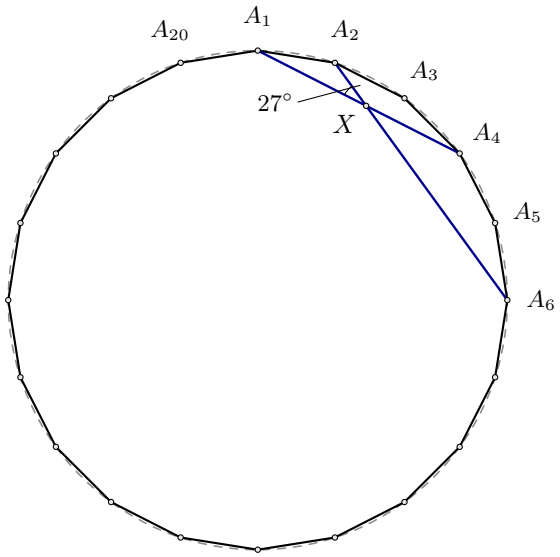
The angles $\angle ACB$ and $\angle ADB$ stand on the same chord AB on the same side of it, so $\angle ACB = 55°$. In triangle BEC the angle at E is $\angle BEC = 180° - 80° = 100°$, so $\angle EBC = 180° - 100° - 55° = 25°$. That is the angle $\angle DBC$.

Finally $\angle CAD$ and $\angle CBD$ stand on the same chord CD, so $\angle CAD = \angle CBD = 25°$.

Answer: $25°$

Problem 5

In a regular n-gon $A_1A_2A_3 \ldots A_n$, the diagonals A_1A_4 and A_2A_6 meet at X, and $\angle A_1XA_2 = 27°$. Find n.



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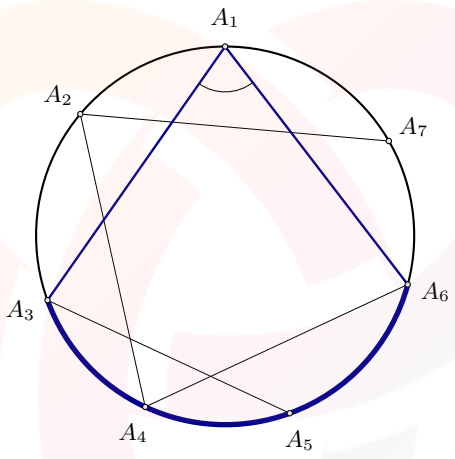
The vertices lie on a circle, and each side subtends $\frac{360^\circ}{n}$ at the centre, hence $\frac{180^\circ}{n}$ at any other vertex. The angle $\angle XA_1A_2 = \angle A_4A_1A_2$ stands on the arc $A_2A_3A_4$, which contains 2 sides, so it equals $2 \times \frac{180^\circ}{n}$. The angle $\angle XA_2A_1 = \angle A_6A_2A_1$ stands on the arc from A_6 round to A_1 that avoids A_2, which contains $n-5$ sides, so it equals $(n-5) \times \frac{180^\circ}{n}$.

In triangle A_1A_2X, therefore, $\angle A_1XA_2 = 180^\circ - (n-3) \times \frac{180^\circ}{n} = \frac{540^\circ}{n}$. Setting $\frac{540}{n} = 27$ gives $n = 20$. (The diagonals do cross, because A_2 lies between A_1 and A_4 while A_6 lies beyond A_4.)

Answer: 20

Problem 6

Seven points $A_1, A_2, \ldots, A_7$ lie on a circle in that order, not necessarily equally spaced. Each point is joined to the two points two places away from it round the circle, so A_1 is joined to A_3 and to A_6, and so on. This draws a seven-pointed star. Find the sum of the seven angles at the tips of the star, $\angle A_6A_1A_3 + \angle A_7A_2A_4 + \angle A_1A_3A_5 + \cdots + \angle A_5A_7A_2$.



Let the arcs $A_1A_2, A_2A_3, \ldots, A_7A_1$ (the short arcs between neighbours) subtend angles $a_1, a_2, \ldots, a_7$ at the centre, so that $a_1 + a_2 + \cdots + a_7 = 360^\circ$.

The tip angle $\angle A_6A_1A_3$ is an angle at the circumference standing on the arc from A_3 to A_6 that avoids A_1. That arc is made of the three small arcs A_3A_4, A_4A_5, A_5A_6, so by the angle at the centre theorem $\angle A_6A_1A_3 = \frac{1}{2}(a_3 + a_4 + a_5)$. In the same way, every tip angle is half the sum of three consecutive a_i, and as we go round the seven tips each a_i is used in exactly three of them.

So the total is $\frac{1}{2} \times 3 \times (a_1 + \cdots + a_7) = \frac{1}{2} \times 3 \times 360^\circ = 540^\circ$. Note that the answer does not depend on where the points are, which is consistent with the regular star, where each tip is $\frac{540^\circ}{7}$.

Answer: 540°